

AMENDMENT NO. 1 MARCH 2024
TO
IS 784 : 2019 PRESTRESSED CONCRETE PIPES (INCLUDING SPECIALS) —
SPECIFICATION
(Third Revision)

(Page 2, clause 5.1, Table 1) — Add the following Note at the end:

‘4 In case of cylinder pipe, minimum core thickness shall be considered including thickness of cylinder’.

[Page 3, clause 7.4.1, Sl No. (i), col (4)] — Substitute the following for the existing:

‘Minimum residual compression of 2.5 N/mm² for up to 500 mm dia and 1.0 N/mm² for 600 mm dia and above’.

[Page 4, Clause 7.4.3, Sl No. (v)] — Substitute the following for the existing:

‘v) Live load condition

Operating condition + $0.13 f_{ck}^{0.67}$ $0.45 f_{ck}$
Live load (with impact)

(Page 4, clause 7.8) — Substitute the following for the existing:

‘The pipes shall be designed to meet the following conditions:

Sl No.	Description of Loading Combination	Permissible Stresses	
		Tension N/mm ²	Compression N/mm ²
(1)	(2)	(3)	(4)
i)	Circumferential prestressing condition (self weight + initial prestress)	0	$0.55 f_p$
ii)	Site test condition (site test pressure + self weight + weight of water + earth fill + final prestress)	$0.38 f_{ck}^{0.67}$	$0.45 f_{ck}$
iii)	Factory test condition (self weight + weight of water + factory test pressure + prestress)	$0.38 f_{ck}^{0.67}$	$0.45 f_{ck}$
iv)	Operating condition (working pressure + self weight + weight of water + earth fill + final prestress)	0	$0.45 f_{ck}$
v)	Live load condition (operating conditions + live load (with impact)	$0.38 f_{ck}^{0.67}$	$0.45 f_{ck}$

where

f_p = characteristic compressive strength of concrete at p days, in N/mm²;

f_{ck} = characteristic compressive design strength of concrete, in N/mm².

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(Page 7, clause 9.1.2.1) — Substitute the following for the existing:

‘Before it is subjected to any further manufacturing process, each steel cylinder with steel cylinder with steel socket and spigot rings welded on it, shall be subjected to water pressure which stresses the steel to at least 140 N/mm² but not more than 175 N/mm² and hydrostatic pressure shall be determined by the following formula:

$$P = \frac{2St_y}{D_{yi}}$$

where

- P = minimum hydrostatic test pressure, in N/mm²;
 S = stress in pipe wall during hydrostatic test, in N/mm², which shall be at least 140 N/mm² and not more than 175 N/mm²;
 t_y = cylinder thickness, in mm; and
 D_{yi} = inside diameter of steel cylinder, in mm.

The test pressure shall be held for 1 min. While under stress, the assembly especially welds, shall be inspected and any parts showing leakage shall be repaired and the whole assembly retested’.

(Page 10, clause 15.1.3.1) — Substitute the following example for the existing:

‘Example:

Suppose the theoretical diameter of socket, $D = 1\,383$ mm

Permissible variation $+2, -0.5$ mm

The maximum diameter, $D_1 = 1\,385$ mm

The minimum diameter, $D_2 = 1\,382.5$ mm

The sway, $S = 2 \times [h \times (D_1 - h)]^{0.5}$
Assume, $h = 5$ mm (see Fig. 1)
Sway, $S = 2 \times [5 \times (1\,385 - 5)]^{0.5}$
 $= 2 \times 83$
 $= 166$ mm

$$\begin{aligned} L^2 &= (S/2)^2 + (D_1 - h)^2 \\ L^2 &= (166/2)^2 + (1\,385 - 5)^2 \\ L^2 &= 6\,889 + 1\,90\,4400 \\ L &= 1\,382.5 \text{ mm} \end{aligned}$$

(Page 20, clause G-4.9) — Substitute the following for the existing:

‘G-4.9 Live Load (W_1) (see 9 of IS 783)’.

(Page 20, clause G-4.10) — Substitute the following for the existing:

‘G-4.10 Calculations of Stresses due to External Loads (see Annex E)’.

(Page 22, clause G-5) — Substitute the word ‘PRESTRESSING’ for ‘PRESTRESS’.

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[Page 23, Table 5, Sl No. (iii) and (iv)] — Substitute the following for the existing:

iii)	Factory test condition (self weight + weight of water + factory test pressure + prestress at factory test)	- (0.464 4 + 0.659 9 + 9.437 2) + 9.940 5 = - 0.621 0	$0.13 (f_{ck})^{0.67}$ = $0.13 \times (40)^{0.67}$ = -1.54	(0.503 8 + 0.593 5 + 9.940 5) = + 11.037 8	$0.45 \times f_{ck}$ = 0.45×40 = 18.0
		- 0.621 0 < - 1.54		+ 11.037 8 < 18.0	
iv)	Operating condition (self weight + weight of water + earth fill load + working pressure + final prestress)	- (0.464 4 + 0.659 9 + 1.567 4 + 5.744 4) + 9.773 1 = + 1.337 0	0.00	(0.503 8 + 0.593 5 + 1.774 2 + 9.731) = + 12.644 6	$= 0.45 \times f_{ck}$ = 0.45×40 = 18.0
		+ 1.337 0 < 0.00		+ 12.644 6 < 18.0	

(Page 27, clause **H-4.8**) — Substitute the following formula for the existing:

‘H-4.8 Weight of Earth Fill, W_e (see Fig. 4 of IS 783)

$$W_e = C_t \times K_e \times (B_t/1\ 000)^2$$

$$W_e = 0.472 \times 15\ 700 \times (1\ 997.20/1\ 000)^2$$

$$W_e = 29\ 558.66\ \text{N/m}$$

where

$$H/B_t = 1.0/1.997\ 2 = 0.500\ 701;$$

$$C_t = 0.472; \text{ and}$$

$$K_e = 15\ 700.'$$

(Page 27, Clause **H-4.9**) — Substitute the following for the existing:

‘H-4.9 Live Load (W_1), (see 9 of IS 783)’.

(Page 27, Clause **H-4.10**) — Substitute the following for the existing:

‘H-4.10 Calculations of Stresses due to External Loads (see Annex E)’.

(Page 28, Clause **H-5**) — Substitute the word ‘PRESTRESSING’ for ‘PRESTRESS’.

(Page 29, Annex H) — Insert the following at the end.

‘H-6 CALCULATION OF HYDROSTATIC TEST PRESSURE OF CYLINDER

As per **9.1.2.1**, each steel cylinder with steel socket and spigot rings welded on it, shall be subjected to water pressure which stresses the steel at least 140 N/mm² but not more than 175 N/mm².

$$P = \frac{2St_y}{D_{yi}}$$

where

$$P = \text{minimum hydrostatic test pressure, in N/mm}^2;$$

$$S = \text{stress in pipe wall during hydrostatic test, in N/mm}^2, \text{ which shall be at least } 140\ \text{N/mm}^2 \text{ and not more than } 175\ \text{N/mm}^2;$$

$$S_{Min} = 140\ \text{N/mm}^2;$$

$$S_{Max} = 175\ \text{N/mm}^2;$$

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t_y = cylinder thickness, in mm = 1.6 mm; and
 D_{yi} = inside diameter of steel cylinder, in mm = 1 350 mm.

Therefore, hydrostatics test pressure for steel cylinder:

$$P_{Min} = \frac{2 \times 140 \times 1.6}{1\,350} = 0.331\,9\text{ N/mm}^2$$

$$P_{Max} = \frac{2 \times 175 \times 1.6}{1\,350} = 0.414\,8\text{ N/mm}^2.$$

(Page 30, *Clause J-2*) — Insert the following at the end:

‘Modulus of elasticity of wire, $E_s = 2\,00\,000\text{ N/mm}^2$ ’.